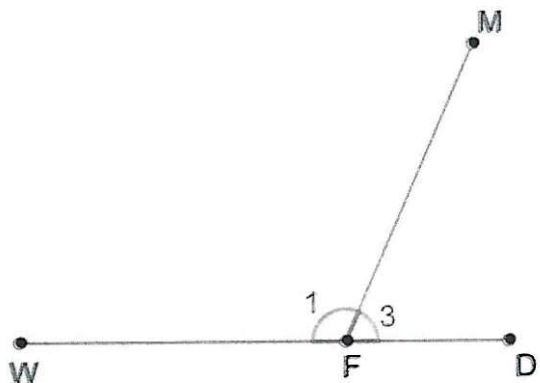
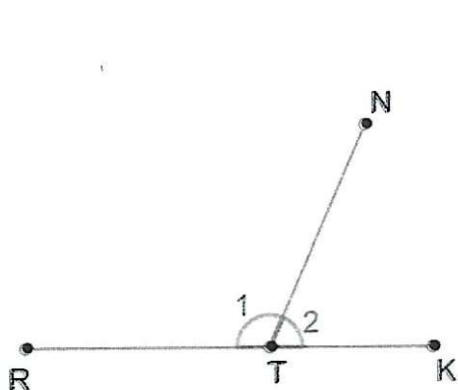


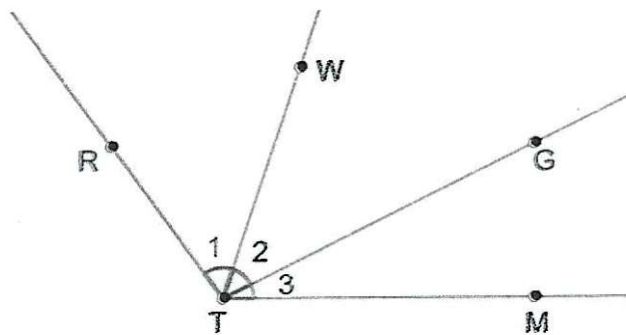
Unit 2: Worked Examples

1. Given $\angle 1$ is supplementary to $\angle 2$ and $\angle 1$ is supplementary to $\angle 3$, prove $\angle 2 \cong \angle 3$ using a two-column proof.



Statement	Reason	Statement	Reason
1) $\angle 1$ is supplementary to $\angle 2$, $\angle 1$ is supplementary to angle 3.	1) given	4) $m\angle 2 = m\angle 3$	4) Subtraction property of equality
2) $m\angle 1 + m\angle 2 = 180^\circ$ $m\angle 1 + m\angle 3 = 180^\circ$	2) Definition of supplementary angles	5) $\angle 2 \cong \angle 3$	5) definition of congruence
3) $m\angle 1 + m\angle 2 = m\angle 1 + m\angle 3$	3) transitive		

1. Given $m\angle 1 = m\angle 3$, prove $\angle RTG \cong \angle WTM$ using a paragraph proof.



It is given that $m\angle 1 = m\angle 3$. By the addition property of equality $m\angle 1 + m\angle 2 = m\angle 3 + m\angle 2$. $m\angle 1 + m\angle 2 = m\angle RTG$ and $m\angle 3 + m\angle 2 = m\angle WTM$ by the angle addition postulate. $m\angle RTG = m\angle WTM$ by the substitution property of equality. $\angle RTG \cong \angle WTM$ by the definition of congruence.